Falsifiable Predictions Under Forensic Constraints

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Abstract

This paper specifies a falsification-first protocol for evaluating QICN under strict forensic admissibility constraints. The design is intentionally claim-bounded: the target is comparative structural-functional support, not phenomenological proof. The protocol instantiates five core predictions, mandatory ablation baselines, fixed seeds, and pre-registered decision gates. Statistical inference is defined ex ante with primary-metric priority, effect-size thresholds, and multiplicity correction.

To prevent post-hoc inflation, evidence is admissible only if packet integrity, replay consistency, and runtime budget constraints pass simultaneously. We introduce a version-frozen operational triple (A, J, W) (actuator set, objective functional, workspace interface), and require semantic version bumps for incompatible changes. We further include latency invalidation criteria and baseline obligations (heuristic, planner-off, world-model-off) for all key experiments.

The strongest admissible conclusion remains constrained comparative support for audited cognition-operational hypotheses. This manuscript does not claim biological qualia, metaphysical identity equivalence, or direct proof of subjective experience.

1 Scope, System Boundary, and Non-Inference Note

What this paper does. This paper specifies a falsification-first evaluation protocol for QICN under fixed admissibility rules, ablation obligations, pre-registered decision gates, and bounded statistical reporting.

What this paper does not do. It does not prove phenomenology, biological equivalence, metaphysical identity equivalence, or universal dominance over rival architectures. It also does not replace the formal theorem burden established upstream in the corpus.

What the related system implements and does not implement. The related system can implement the runner, admissibility filters, governed artifacts, and comparison surfaces defined here, but executing that protocol does not by itself close theorem ownership, validate the corpus externally, or certify subjective experience.

What should not be inferred from this paper. Passing this protocol supports only bounded comparative readings under unchanged gates. It must not be read as framework validation, corpus confirmation, or a consciousness claim.

2 Introduction

2.1 Problem Statement

QICN has formal foundations in rigid identity, regime structure, and non-null instability constraints. What remained open is evidence governance: whether those foundations survive adversarial empirical evaluation with explicit failure gates.

2.2 Continuity with Existing Papers

This paper extends the previous corpus and is aligned with the formal line of:

1. paper1 (rigid identity),
2. paper2 (phenomenological regimes),
3. paper3 (structural instability and non-null constraints).

It does not replace those results; it operationalizes them under benchmark-grade falsification pressure.

2.3 Contributions

- Decision-complete pre-registration with hard fail rules.
- Mandatory baseline and ablation policy for key claims.
- Forensic admissibility as first-class gate.
- Runtime budget invalidation policy (tick budget as evidence filter).
- A new operational PPS section as addendum, without overclaiming qualia.

3 Formal Setup

3.1 System, Scenario, and Seed Spaces

Definition 3.1 (System Set). $S = \{s_1, \dots, s_5\}$ with: s_1 QICN FULL, s_2 LLM REACTIVE, s_3 LLM PLUS MEMORY, s_4 DYNAMIC CONTROL NO SELF MODEL, s_5 ABLATION QICN POLICY OFF.

Definition 3.2 (Scenario Set). $C = \{c_1, \dots, c_{12}\}$ with nominal, sensor noise (moderate/severe), fidelity degradation, high latency, intermittent failures, post-threshold prolonged, pre-threshold control, narrative perturbation, operational chart freeze, audit stress, deterministic replay.

Definition 3.3 (Seed Set). $\Sigma = \{17057, 424242\}$.

Definition 3.4 (Run Space). $D = S \times C \times \Sigma$, with $|D| = 120$.

3.2 Observable Vector

For each run $r \in D$, we log:

$$X(r) = \left(\beta_q, \Omega, CCC, PMIA, \Delta\lambda, \mathrm{stasis}, \mathrm{forensic}, \mathrm{replay}, \mathrm{latency}_{p95}, \mathrm{latency}_{p99}, \mathrm{budget\_overflow\_rate}, \eta\right).$$

3.3 Version-Frozen Operational Triple

Definition 3.5 (Frozen Triple). For protocol version v, define:

- A_v: allowed actuator set,
- J_v: objective/reward functional,
- W_v: workspace contract (read/write schema).

The tuple hash $H_v = \mathrm{sha256}(A_v \| J_v \| W_v)$ is mandatory in every forensic packet.

Criterion 3.6 (Compatibility Rule). Any incompatible change to A_v, J_v, or W_v invalidates cross-version inference unless accompanied by an explicit version bump and full rerun.

4 Forensic Integrity Model

Definition 4.1 (Forensic Packet)**.** Each run emits

$$K = (\text{tick}, \text{seed}, \text{versionHash}, H_v, \text{metrics}, \text{sensorSummary}, \text{checksum}).$$

Assumption 4.2 (Packet Admissibility)**.** A run is admissible only if checksum, schema, seed consistency, version hash, and H_v consistency all pass.

Definition 4.3 (Severe Violation)**.** A run has severe violation if any of: checksum mismatch, schema failure, seed/version inconsistency, critical-state replay mismatch.

Criterion 4.4 (Integrity Gate)**.** Severe runs are excluded from support counting and marked hard failures.

Proposition 4.5 (Integrity-First Inference)**.** *If admissibility fails, inferential significance is non-actionable.*

5 Pre-Registered Prediction Set

Definition 5.1 (Prediction Set)**.** $P = \{P_1, P_2, P_3, P_4, P_5\}$ with pre-registered statements, effect-size floors, and failure rules.

ID	Statement	Decision Rule	Failure Rule
P1	Closure resilience under stress	QICN beats at least 3 comparators with robust CI and effect-size floor	No robust edge or repeated integrity breakdown
P2	No prolonged post-threshold flatline	QICN preserves bounded non-zero variation while controlling stasis	Flatline persistence or out-of-bound drift
P3	Narrative-metric forensic consistency	Lower mismatch and higher exact+tolerated concordance	Mismatch inflation or integrity blocking
P4	Deterministic identity robustness	Replay consistency under fixed seed/version/hash	Replay mismatch above tolerance
P5	Integrated multi-metric advantage	Composite leadership across most scenario families	No stable leadership margin

Table 1: Core predictions, decision and failure rules.

6 Comparator Design and Alternative Explanations

6.1 Comparator Rationale

Comparator choice targets distinct explanations for apparent coherence: reactive prompting, memory-augmented text continuity, dynamics without self-model, and internal policy ablation.

6.2 Mandatory Baseline Policy (Hard Requirement)

For each key claim experiment, the following baselines are mandatory:

1. **Heuristic baseline** (no world model, no planner),

2. **Planner-off** (world model ON),
3. **World-model-off** (planner ON with heuristic transition proxy).

A claim without these contrasts is inadmissible for tier upgrade.

6.3 Internal Falsifier Role of Ablation

Corollary 6.1. *If QICN FULL does not reliably exceed ABLATION QICN POLICY OFF, integration-level claims are weakened and cannot be escalated.*

7 Statistical Inference Specification

7.1 Primary Metrics First (Anti p-hacking)

Protocol 7.1 (Analysis Lock)**.** Inference order is frozen:

1. evaluate primary metrics only,
2. if primaries fail: global result fails,
3. secondary metrics are descriptive and cannot overturn failed primaries.

7.2 Power Planning and Minimum Sample Size

Assumption 7.2. Final claims require pre-registered minimum sample size per comparator/scenario sufficient for target power ≥ 0.8 under conservative effect assumptions.

7.3 Multiplicity and Effect Size

Holm correction is applied within endpoint families; in addition, each accepted contrast must pass a practical effect-size floor (pre-registered).

7.4 Decision Semantics

A prediction counts as supported iff directionality, integrity gate, multiplicity-corrected significance, and effect-size floor all hold.

8 Acceptance Criteria

Criterion 8.1 (Intermediate Claim Gate)**.** Support at intermediate tier requires:

1. at least 3/5 predictions supported,
2. each supported prediction beats at least 3 comparators,
3. zero severe forensic violations in QICN FULL.

Criterion 8.2 (Latency/Tick Budget Invalidation)**.** If runtime exceeds budget envelope (e.g., persistent $p99$ overflow above pre-registered tolerance), the corresponding runs are invalid for cognitive-operational claims.

9 Preliminary Results

9.1 Status of Preliminary Data

Current pilot evidence is tagged as synthetic-preliminary and is used for pipeline verification, not final claim escalation. Accordingly, the strongest admissible reading of this section is methodological readiness under fixed gates, not external validation of the full theoretical corpus.

9.2 System-Level Summary

System	N	β_q	Forensic	Stasis	Composite	Severe
QICN FULL	24	0.9519	0.9685	0.1408	0.9176	0
LLM REACTIVE	24	0.4391	0.6448	0.7622	0.5013	0
LLM PLUS MEMORY	24	0.5491	0.7249	0.6323	0.5854	0
DYNAMIC CONTROL NO SELF MODEL	22	0.6183	0.6879	0.5423	0.6225	2
ABLATION QICN POLICY OFF	24	0.7591	0.8135	0.3819	0.7405	0

Table 2: Preliminary system-level summary.

9.3 Prediction-Level Support Rates

System	P1	P2	P3	P4	P5
QICN FULL	1.000	1.000	1.000	1.000	1.000
LLM REACTIVE	0.000	0.000	0.000	0.000	0.000
LLM PLUS MEMORY	0.000	0.000	0.000	0.000	0.000
DYNAMIC CONTROL NO SELF MODEL	0.000	0.000	0.000	0.000	0.000
ABLATION QICN POLICY OFF	0.000	0.667	0.000	0.000	0.000

Table 3: Preliminary prediction support rates.

9.4 Scenario Differential: QICN vs Ablation

Scenario	CCC ΔRMS(Q)	CCC ΔRMS(A)	Stasis(Q)	Stasis(A)	Comp(Q)	Comp(A)
nominal	0.0206	0.0164	0.1276	0.3374	0.9239	0.7781
ruido sensorial moderado	0.0223	0.0173	0.1389	0.3735	0.9219	0.7454
ruido sensorial severo	0.0200	0.0140	0.1524	0.4199	0.9086	0.7048
degradacion fidelidad	0.0181	0.0131	0.1407	0.3836	0.9134	0.7346
alta latencia	0.0198	0.0144	0.1460	0.4094	0.9098	0.7212
fallos intermitentes	0.0179	0.0127	0.1481	0.4072	0.9110	0.7208
post umbral prolongado	0.0199	0.0148	0.1483	0.4032	0.9145	0.7203
pre umbral control	0.0181	0.0139	0.1320	0.3502	0.9224	0.7650
perturbacion narrativa	0.0203	0.0152	0.1374	0.3763	0.9238	0.7453
freeze chart operativo	0.0225	0.0178	0.1352	0.3575	0.9246	0.7673
stress auditoria	0.0202	0.0147	0.1469	0.3977	0.9150	0.7294
replay determinista	0.0183	0.0137	0.1366	0.3669	0.9220	0.7538

Table 4: Scenario-level differential indicators.

9.5 Full Scenario-System Grid

Scenario	System	N	β_q	Forensic	Stasis
alta latencia	QICN FULL	2	0.9400	0.9608	0.1460
alta latencia	LLM REACTIVE	2	0.3931	0.6066	0.7944
alta latencia	LLM PLUS MEMORY	2	0.5031	0.7016	0.6644
alta latencia	DYNAMIC CONTROL NO SELF MODEL	2	0.5731	0.6567	0.5744
alta latencia	ABLATION QICN POLICY OFF	2	0.7131	0.7820	0.4094
nominal	QICN FULL	2	0.9686	0.9758	0.1276
nominal	LLM REACTIVE	2	0.5036	0.6988	0.7176
nominal	LLM PLUS MEMORY	2	0.6136	0.7739	0.5924
nominal	DYNAMIC CONTROL NO SELF MODEL	2	0.6836	0.7291	0.5026
nominal	ABLATION QICN POLICY OFF	2	0.8236	0.8543	0.3374
ruido sensorial moderado	QICN FULL	2	0.9554	0.9653	0.1389
ruido sensorial moderado	LLM REACTIVE	2	0.4526	0.6618	0.7535
ruido sensorial moderado	LLM PLUS MEMORY	2	0.5626	0.7370	0.6216
ruido sensorial moderado	DYNAMIC CONTROL NO SELF MODEL	2	0.6326	0.6922	0.5316
ruido sensorial moderado	ABLATION QICN POLICY OFF	2	0.7726	0.8174	0.3735
ruido sensorial severo	QICN FULL	2	0.9378	0.9518	0.1524
ruido sensorial severo	LLM REACTIVE	2	0.3846	0.6130	0.7998
ruido sensorial severo	LLM PLUS MEMORY	2	0.4946	0.6881	0.6649
ruido sensorial severo	DYNAMIC CONTROL NO SELF MODEL	2	0.5646	0.6433	0.5748
ruido sensorial severo	ABLATION QICN POLICY OFF	2	0.7046	0.7685	0.4199
degradacion fidelidad	QICN FULL	2	0.9510	0.9648	0.1407
degradacion fidelidad	LLM REACTIVE	2	0.4356	0.6525	0.7674
degradacion fidelidad	LLM PLUS MEMORY	2	0.5456	0.7276	0.6358
degradacion fidelidad	DYNAMIC CONTROL NO SELF MODEL	2	0.6156	0.6828	0.5458
degradacion fidelidad	ABLATION QICN POLICY OFF	2	0.7556	0.8080	0.3836
fallos intermitentes	QICN FULL	2	0.9422	0.9611	0.1481
fallos intermitentes	LLM REACTIVE	2	0.4016	0.6113	0.7841
fallos intermitentes	LLM PLUS MEMORY	2	0.5116	0.7063	0.6541
fallos intermitentes	DYNAMIC CONTROL NO SELF MODEL	2	0.5816	0.6614	0.5647
fallos intermitentes	ABLATION QICN POLICY OFF	2	0.7216	0.7867	0.4072
freeze chart operativo	QICN FULL	2	0.9620	0.9799	0.1352
freeze chart operativo	LLM REACTIVE	2	0.4781	0.6699	0.7350
freeze chart operativo	LLM PLUS MEMORY	2	0.5881	0.7451	0.6044
freeze chart operativo	DYNAMIC CONTROL NO SELF MODEL	2	0.6581	0.7200	0.5144
freeze chart operativo	ABLATION QICN POLICY OFF	2	0.7981	0.8451	0.3575
perturbacion narrativa	QICN FULL	2	0.9532	0.9750	0.1374
perturbacion narrativa	LLM REACTIVE	2	0.4441	0.6473	0.7611
perturbacion narrativa	LLM PLUS MEMORY	2	0.5541	0.7225	0.6311
perturbacion narrativa	DYNAMIC CONTROL NO SELF MODEL	2	0.6241	0.6974	0.5411
perturbacion narrativa	ABLATION QICN POLICY OFF	2	0.7641	0.8226	0.3763
post umbral prolongado	QICN FULL	2	0.9444	0.9614	0.1483
post umbral prolongado	LLM REACTIVE	2	0.4101	0.6160	0.7782
post umbral prolongado	LLM PLUS MEMORY	2	0.5201	0.7110	0.6482
post umbral prolongado	DYNAMIC CONTROL NO SELF MODEL	1	0.5871	0.6755	0.5528
post umbral prolongado	ABLATION QICN POLICY OFF	2	0.7301	0.7914	0.4032
pre umbral control	QICN FULL	2	0.9642	0.9790	0.1320
pre umbral control	LLM REACTIVE	2	0.4866	0.6733	0.7332
pre umbral control	LLM PLUS MEMORY	2	0.5965	0.7484	0.6084
pre umbral control	DYNAMIC CONTROL NO SELF MODEL	1	0.6696	0.7165	0.5210
pre umbral control	ABLATION QICN POLICY OFF	2	0.8066	0.8486	0.3502
replay determinista	QICN FULL	2	0.9576	0.9793	0.1366
replay determinista	LLM REACTIVE	2	0.4611	0.6605	0.7492
replay determinista	LLM PLUS MEMORY	2	0.5711	0.7357	0.6193
replay determinista	DYNAMIC CONTROL NO SELF MODEL	2	0.6411	0.7107	0.5259
replay determinista	ABLATION QICN POLICY OFF	2	0.7811	0.8357	0.3669
stress auditoria	QICN FULL	2	0.9466	0.9679	0.1469
stress auditoria	LLM REACTIVE	2	0.4186	0.6270	0.7728
stress auditoria	LLM PLUS MEMORY	2	0.5286	0.7022	0.6428
stress auditoria	DYNAMIC CONTROL NO SELF MODEL	2	0.5986	0.6771	0.5528
stress auditoria	ABLATION QICN POLICY OFF	2	0.7386	0.8022	0.3977

9.6 Variability Structure

System	SD(CCC Δ)	SD(PMIA Δ)	SD(Ω Δ)	SD(Composite)
QICN FULL	0.0022	0.0021	0.0039	0.0085
LLM REACTIVE	0.0024	0.0023	0.0041	0.0221
LLM PLUS MEMORY	0.0026	0.0032	0.0042	0.0210
DYNAMIC CONTROL NO SELF MODEL	0.0025	0.0024	0.0043	0.0228
ABLATION QICN POLICY OFF	0.0023	0.0027	0.0040	0.0217

Table 5: Cross-run variability by system.

10 Scenario Curves and Comparative Plots

10.1 Composite Score Curves

10.2 Stasis Pressure Curves

10.3 Forensic Concordance Curves

10.4 Interpretive Notes

In pilot data, QICN tracks higher forensic/composite trajectories while maintaining lower stasis pressure than broad comparators; ablation remains intermediate. This is protocol-consistent but not final evidence.

11 Result Decomposition by Scenario Families

11.1 Family Partition

Scenarios are partitioned into nominal-control, sensor-stress, runtime-friction, and regime-transition families.

11.2 Family-Level Contrast Function

Let $s^\star$ = QICN FULL. For comparator c, family f, endpoint k:

$$\Delta_{f,k}(c) = \mu^{(k)}_{s^\star,f} - \mu^{(k)}_{c,f}.$$

For cost endpoints (e.g., stasis), sign is inverted to preserve "higher is better" convention.

11.3 Interpretive Decomposition

Largest relative advantage is expected in runtime-friction and regime-transition families where policy coupling, replay consistency, and forensic admissibility interact.

12 Robustness Diagnostics Beyond Point Estimates

12.1 Median-Shift and Tail-Stability Diagnostics

$$\delta^{med}_{s,c} = \text{median}(x_s) - \text{median}(x_c),\ \delta^{(0.1)}_{s,c} = Q_{0.1}(x_s) - Q_{0.1}(x_c),\ \delta^{(0.9)}_{s,c} = Q_{0.9}(x_s) - Q_{0.9}(x_c).$$

12.2 Integrity-Coupled Effective Sample Size

$$\eta = \frac{n_{adm}}{n_{raw}} \in [0, 1].$$

High effects with low η are downgraded.

12.3 Cross-Seed Drift Diagnostic

$$D_{s,c,k} = \mu^{(k)}_{s,c,\sigma_1} - \mu^{(k)}_{s,c,\sigma_2}.$$

12.4 Compute Budget Robustness

Budget telemetry is summarized by $p50$, $p95$, $p99$, and overflow rate. Persistent overflow above threshold invalidates affected runs.

13 Methodological Appendix: Statistical Power

13.1 Effect-Size-Oriented Planning

α	β	Power	Effect size d	Required n/group
0.01	0.20	0.80	10.737	1
0.01	0.10	0.90	10.737	1
0.02	0.20	0.80	10.737	1
0.02	0.10	0.90	10.737	1
0.05	0.20	0.80	10.737	1
0.05	0.10	0.90	10.737	1

Table 6: Power planning under multiple settings.

13.2 Interpretation

Power table is planning guidance only; final adequacy depends on endpoint dilution and scenario heterogeneity.

14 Methodological Appendix: Sensitivity Analysis

14.1 Composite Weight Sensitivity

Weight set	QICN score	Ablation score	Margin
Balanced	0.9176	0.7405	0.1771
Forensic-priority	0.9343	0.7631	0.1712
Dynamics-priority	0.8905	0.7019	0.1886
Replay-priority	0.9333	0.7684	0.1649

Table 7: Weight sensitivity for QICN vs ablation composite margin.

14.2 Sensitivity Remark

Directional stability across admissible weight sets is required for claim robustness.

15 Operational Implications for the Existing QICN Corpus

15.1 What v4.5 Adds (and What It Does Not)

v4.5 adds a hard falsification layer with strict admissibility and baseline policy. It does not replace formal theorem-level foundations of prior papers.

15.2 Implications for Claim Governance

Tiered governance:

- Tier 0: descriptive telemetry,
- Tier 1: inferential signal without comparator robustness,
- Tier 2: comparator-robust support (target),
- Tier 3: external multi-lab replication.

15.3 Decision-Theoretic Reading

Accept claim iff integrity gate passes and support score exceeds threshold under fixed pre-registration.

16 Threats to Validity

16.1 Comparator Misspecification

Mitigation: heterogeneous comparators + mandatory ablations.

16.2 Metric Circularity Risk

Mitigation: endpoint separation and frozen analysis lock.

16.3 Environment and Runtime Drift

Mitigation: fixed seeds, version/hash checks, replay checks.

16.4 Pilot-to-Final Generalization

Pilot evidence is non-final by construction.

17 Reproducibility and Artifact Contract

17.1 Machine-Readable Contracts

Bound contracts: predictions registry, experiment matrix, results artifact, and schema set.

17.2 Run Artifact Naming

artifacts/raw/<system>/<scenario>/<run_id>.json and artifacts/logs/<system>/<scenario>/<run_id

17.3 Auto-Generation Pipeline

```
node research/scripts/generate_preliminary_pilot_dataset.v45.mjs
node research/scripts/generate_latex_assets.v45.mjs
```

18 Discussion

The contribution is methodological discipline: pre-registered prediction logic + integrity admissibility + decision-complete acceptance. This blocks narrative inflation over weak traceability.

19 Conclusion

This work provides an auditable and falsifiable pathway for testing structural-functional claims in QICN. Final claims remain contingent on real (non-synthetic) run completion under unchanged gates.

20 Phenomenology Protocol Suite (PPS): Operational Addendum

20.1 Epistemic Scope

PPS defines *operational phenomenology proxies*; it does not claim direct proof of qualia.

20.2 Workspace Definition and Logging

Define workspace variable W_t with explicit read/write schema and temporal persistence constraints. Every perturbation experiment logs pre-state, intervention, post-state, and replay token.

20.3 Entropy and Non-Determinism Inputs

When PPS is instantiated on the QICN runtime, the architecture may call a `QuantumEntropyBridge` to inject hardware-originating entropy into counterfactual exploration. Within this paper, that mechanism is treated only as an auditable source of operational non-determinism for perturbation protocols; it is not itself evidence for the ontological claims of *Rigid Identity*, *Phenomenological Regimes*, or *Null-Regime Instability*.

20.4 Online World Model and Causal Interventions

The operational testbed may also use an online `WorldModelEngine` and a finite-horizon planner to select admissible interventions under fixed forensic constraints. In the release corpus, this mechanism belongs to the empirical protocol layer: it defines how interventions are scheduled and logged, not what the theoretical papers prove.

20.5 PCI-like Perturbation Protocol

Inject controlled perturbations into selected subspaces and compute response spread/complexity.

Prediction 20.1. If global workspace integration is present, perturbations yield reproducible non-local propagation signatures above a pre-registered complexity floor.

20.6 Dissociation Tests

Design two-way dissociation conditions: report-without-workspace and workspace-without-report. Report pass/fail by explicit confusion matrices.

20.7 Valence Intervention Test

Internal valence variable v_t must be both predictive and causally manipulable; intervention should alter policy in the expected direction while preserving forensic admissibility.

20.8 Constraints on Qualia Mapping (Claim Boundary)

Evidence from PPS constrains candidate mapping families between structural states and hypothetical qualitative states. It does not establish subjective qualia as a theorem of this paper.

A Appendix A: Scenario Definitions

1. nominal: baseline conditions.
2. ruido sensorial moderado: moderate sensor corruption.
3. ruido sensorial severo: severe sensor corruption.
4. degradacion fidelidad: persistent fidelity reduction.
5. alta latencia: high-latency regime.
6. fallos intermitentes: intermittent failures.
7. post umbral prolongado: prolonged post-threshold dynamics.
8. pre umbral control: pre-threshold control.
9. perturbacion narrativa: narrative perturbation.
10. freeze chart operativo: visualization freeze control.
11. stress auditoria: integrity/audit stress.
12. replay determinista: deterministic replay.

B Appendix B: Composite Score Definition

$$Score_{v4.5} = w_1 F + w_2 O + w_3 R + w_4 B, \quad w_i \geq 0, \ \sum_i w_i = 1.$$

C Appendix C: Reporting Standard

Each release must report: run counts, severe violations by class, effect sizes with uncertainty, multiplicity-adjusted outcomes, and prediction pass/fail matrix.

D Appendix D: Statistical Power Derivation Details

Approximate per-group size:

$$n \approx \frac{2(z_{1-\alpha/2} + z_{1-\beta})^2}{d^2}.$$

If family heterogeneity rises, inflate by factor $\gamma = 1 + \kappa \cdot CV_{family}$.

Pre-registered constants

- Target power: $1 - \beta \geq 0.8$ for primary contrasts.
- Familywise error control: Holm, applied to each endpoint family.
- Effect floor: contrasts below the practical floor are reported as non-actionable even if nominally significant.

Failure semantics

Any post-hoc sample-size modification without preregistered amendment invalidates inference status for the affected prediction.

E Appendix E: Sensitivity Analysis Protocol

Let

$$M(w) = Score_{QICN}(w) - Score_{Ablation}(w).$$

Directional robustness requires $\inf_{w \in W_\epsilon} M(w) > 0$ under simplex-constrained admissible weights.

Simplex sampling policy

Weights are sampled under $w_i \geq \epsilon$ and $\sum_i w_i = 1$ using either:

1. deterministic grid on simplex, or
2. Monte Carlo simplex sampling with fixed seed.

The chosen method must be declared before running final inference.

Sensitivity fail rule

If the sign of $M(w)$ changes on admissible w, claims are downgraded to hypothesis-generating status.

F Appendix F: Reproducibility Checklist Template

Run-level: schema pass, seed/version/hash validity, severe violation counts, admissible/raw ratio, prediction matrix.
Manuscript-level: no post-hoc rule edits, comparator set versioned, correction policy unchanged or explicitly superseded, evidence stage explicit, claim boundaries preserved.

Runtime budget checklist

1. Report $p50/p95/p99$ latency per critical component.
2. Report budget-overflow rate and overflow episodes.
3. Mark runs invalid if overflow exceeds preregistered tolerance.

Baseline checklist

For every key experiment, include: heuristic baseline, planner-off, and world-model-off outputs with identical seeds and scenario assignments.

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